

Pumping Lemma In Class Examples and Extra Problems

Use the pumping lemma to show that the following languages are not regular.

1. Let $\Sigma = \{a, b\}$. $L = \{w \in \Sigma^* : n_a(w) < n_b(w)\}$

- Assume by contradiction that L is regular.
- Pumping Lemma states there exists an integer $m > 0$ such that $\forall w \in L$, w can be decomposed $w = xyz$ with $|xy| \leq m$ and $|y| > 0$ and $w_i = xy^iz \in L$.
- Choose our string w . $w = xyz$. Since $|xy| \leq m$ and $|y| \geq 1 \Rightarrow y$ contains only a's $y = a^k$ with $k \geq 1$

$$w = a^m b^{2m}$$

$$w_i = a^{m-k}(a^k)^i b^{2m} \in L$$

- Choose i . $i = 2$ $w_2 = a^{m-k}(a^k)^2 b^{2m} = a^{m-k} a^{2k} b^{2m} = a^{m-k+2k} b^{2m} = a^{m+k} b^{2m} \in L$

Let's try another value for i . $i = m$ $w_m = a^{m-k}(a^k)^m b^{2m} = a^{m-k} a^{mk} b^{2m} = a^{m-k+mk} b^{2m} = a^{m(1+k)-k} b^{2m} \in L$

These strings that are pumped are part of L . We will need to choose another string w . $w = xyz$. Since $|xy| \leq m$ and $|y| \geq 1 \Rightarrow y$ contains only a's $y = a^k$ with $k \geq 1$

$$w = a^m b^{m+1}$$

$$w_i = a^{m-k}(a^k)^i b^{m+1} \in L$$

$$w_2 = a^{m-k}(a^k)^2 b^{m+1} = a^{m-k} a^{2k} b^{m+1} = a^{m+k} b^{m+1} \notin L$$

This is a contradiction from our assumption. Therefore, L is not regular.

We could also use the string $w = b^m a^{m-1}$. The pumping lemma would generate $w_i = b^{m-k}(b^k)^i a^{m-1}$ where $k \geq 1$

$$w_i = b^{m-k} a^{m-1} \in L$$

$$w_0 = b^{m-k} a^{m-1} \notin L$$

This is another contradiction from our assumption. Therefore, L is not regular.

$$2. L = \{(ab)^n a^k : n > k, k \geq 0\}$$

- Assume by contradiction that L is regular.
- Pumping Lemma states there exists an integer $m > 0$ such that $\forall w \in L$, w can be decomposed $w = xyz$ with $|xy| \leq m$ and $|y| > 0$ and $w_i = xy^i z \in L$.
- Choose our string w . $w = xyz$. Since $|xy| \leq m$ and $|y| \geq 1 \Rightarrow y$ contains a pair $ab = ab$ with $k \geq 1$.

$$w = (ab)^{m+1} a^m = (ab)^m aba^m$$

$$w_i = (ab)^{m-k} ((ab)^k)^i aba^m \in L$$

- Choose i . $i = 0$, this would give

$$w_0 = (ab)^{m-k} aba^m = (ab)^{m-k+1} a^m \notin L$$

This is a contradiction from our assumption. Therefore, L is not regular.

$$3. L = \{a^n : n \text{ is a perfect square}\}$$

- Assume by contradiction that L is regular.
- Pumping Lemma states there exists an integer $m > 0$ such that $\forall w \in L$, w can be decomposed $w = xyz$ with $|xy| \leq m$ and $|y| > 0$ and $w_i = xy^i z \in L$.
- Choose our string w . $w = xyz$. Since $|xy| \leq m$ and $|y| \geq 1$.

$$w = a^{m^2}$$

From w we can decompose it where x , y , and z contains some number of a 's.
Where

- $x = a^l$
- $y = a^k \quad k \geq 1$
- $z = a^{m^2-l-k}$

$$w = a^l a^k a^{m^2-l-k} \in L$$

$$w_i = a^l (a^k)^i a^{m^2-l-k} \in L$$

- Choose i . $i = 2$, this would give

$$w_2 = a^l (a^{2k}) a^{m^2-l-k} = a^{m^2+k} \notin L$$

We know that $m^2 < m^2 + k \leq m^2 + m < (m+1)^2$

For the first part of the inequality $m^2 < m^2 + k$ since $k \geq 1$, The second part of the inequality $m^2 + k \leq m^2 + m$ is true since the decomposition $|xy| \leq m$ and $|y| \leq m$. The third part of the inequality looks at the next perfect square $(m+1)^2$ which is strictly greater than $m^2 + m$ since $(m+1)^2 = m^2 + 2m + 1$. The plus 1 causes the strict inequality relation. This causes the idea that there is no perfect square between the two consecutive perfect squares since $m^2 + k$ is not ever a perfect square.

This is a contradiction from our assumption. Therefore, L is not regular.

4. $L = \{a^n b^k c^{n+k} \mid n \geq 0, k \geq 0\}$

- Assume by contradiction that L is regular.
- Pumping Lemma states there exists an integer $m > 0$ such that $\forall w \in L$, w can be decomposed $w = xyz$ with $|xy| \leq m$ and $|y| > 0$ and $w_i = xy^i z \in L$.
- Choose our string w . $w = xyz$. Since $|xy| \leq m$ and $|y| \geq 1$.

$$w = a^m b^m c^{2m}$$

From w we can decompose it where $y = a^k \mid k > 0$

$$w_i = a^{m-k} (a^k)^i b^m c^{2m} \in L$$

- Choose i . $i = 0$, this would give

$$w_0 = a^{m-k} b^m c^{2m} \notin L$$

This is a contradiction from our assumption. Therefore, L is not regular.

5. $L = \{a^n b^l a^k \mid k \leq n + l\}$

- Assume by contradiction that L is regular.
- Pumping Lemma states there exists an integer $m > 0$ such that $\forall w \in L$, w can be decomposed $w = xyz$ with $|xy| \leq m$ and $|y| > 0$ and $w_i = xy^i z \in L$.
- Choose our string w . $w = xyz$. Since $|xy| \leq m$ and $|y| \geq 1$.

$$w = a^m b^m a^{2m}$$

From w we can decompose it where $y = a^k \mid k > 0$

$$w_i = a^{m-k} (a^k)^i b^m a^{2m} \in L$$

- Choose i . $i = 0$, this would give

$$w_0 = a^{m-k} b^m a^{2m} \notin L$$

This is a contradiction from our assumption. Therefore, L is not regular.

6. $L = \{ww : w \in \{a,b\}^*\}$

- Assume by contradiction that L is regular.
- Pumping Lemma states there exists an integer $m > 0$ such that $\forall w \in L$, w can be decomposed $w = xyz$ with $|xy| \leq m$ and $|y| > 0$ and $w_i = xy^iz \in L$.
- Choose our string w . $w = xyz$. Since $|xy| \leq m$ and $|y| \geq 1$.

$$w = a^m b a^m b$$

From w we can decompose it where $y = a^k$ $k > 0$

$$w_i = a^{m-k} (a^k)^i b a^m b \in L$$

- Choose i . $i = 0$, this would give

$$w_0 = a^{m-k} b a^m b \notin L$$

This is a contradiction from our assumption. Therefore, L is not regular.

7. $L = \{w\bar{w} : w \in \{0,1\}^*\}$

- Assume by contradiction that L is regular.
- Pumping Lemma states there exists an integer $m > 0$ such that $\forall w \in L$, w can be decomposed $w = xyz$ with $|xy| \leq m$ and $|y| > 0$ and $w_i = xy^iz \in L$.
- Choose our string w . $w = xyz$. Since $|xy| \leq m$ and $|y| \geq 1$.

$$w = 001^m \text{ and } \bar{w} = 110^m \quad m > 0$$

$$w\bar{w} = 001^m 110^m = 001^{m-2} 11110^m$$

From w we can decompose it where

$$y = 1^{(m-2)k} \quad k > 0$$

$$w_i = 00(1^{mk-2k})^i 11110^m \in L$$

- Choose i . $i = 0$, this would give

$$w_0 = 0011110^m \notin L$$

This is a contradiction from our assumption. Therefore, L is not regular. Note: You could also choose $w = 0^m$ which would lead to $w\bar{w} = 0^m 1^m$. This is a similar example string that you have worked on previously.

8. $L = \{a^n b^l : n \leq l\}$

- Assume by contradiction that L is regular.
- Pumping Lemma states there exists an integer $m > 0$ such that $\forall w \in L$, w can be decomposed $w = xyz$ with $|xy| \leq m$ and $|y| > 0$ and $w_i = xy^i z \in L$.
- Choose our string w . $w = xyz$. Since $|xy| \leq m$ and $|y| \geq 1$.

$$w = a^m b^m$$

From w we can decompose it where $y = a^k$ $k > 0$

$$w_i = a^{m-k}(a^k)^i b^m \in L$$

- Choose i . $i = 2$, this would give

$$w_2 = a^{m-k}(a^k)^2 b^m = a^{m-k+2k} b^m = a^{m+k} b^m \notin L$$

This is a contradiction from our assumption. Therefore, L is not regular.

9. $L = \{a^n b^l a^k : n = l \text{ or } l \neq k\}$

- Assume by contradiction that L is regular.
- Pumping Lemma states there exists an integer $m > 0$ such that $\forall w \in L$, w can be decomposed $w = xyz$ with $|xy| \leq m$ and $|y| > 0$ and $w_i = xy^i z \in L$.
- Choose our string w . $w = xyz$. Since $|xy| \leq m$ and $|y| \geq 1$.

$$w = a^m b^m a^m$$

From w we can decompose it where $y = a^k$ $k > 0$

$$w_i = a^{m-k}(a^k)^i b^m a^m \in L$$

- Choose i . $i = 0$, this would give

$$w_0 = a^{m-k} b^m a^m \notin L$$

This is a contradiction from our assumption. Therefore, L is not regular.